Simulation Study: Base Model and Sensitivity Scenarios

A Bayesian Framework for Causal Inference with Error-Prone Environmental Exposures

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1 Motivation

The simulation reflects the two-level structure of the application: area-level count health outcomes for participants nested within Forward Sortation Areas (FSA or the first three characters of a Canadian postal code), with area-level exposure assigned from a spatiotemporal prediction model.

2 Notation

- $i = 1, \ldots, I$: area index (FSAs)
- A_i: latent true area-level $PM_{2.5}$ exposure, unobserved
- Z_i: observed error-prone surrogate from a spatiotemporal prediction model (e.g. CanOSSEM)
- $\mathbf{X}_i = (X_{i1}, \ldots, X_{i4})^\top$: area-level confounders affecting both exposure and outcome
- D_i: area-level calibration covariate explaining prediction error magnitude in Z_i (e.g. distance to nearest monitoring station, area deprivation)
- Y_i: area-level count health outcome (number of incident depression cases)
- N_i: area population size, used as Poisson offset
- ϕ_i: spatial random effect on the latent exposure surface, following the CAR prior in equation (1)
- $\mathbf{W}$: $I \times I$ binary adjacency matrix ($W_{ij} = 1$ if areas i and j share a border)
- $\mathbf{D}_W = \text{diag}(\mathbf{W1})$: diagonal degree matrix (number of neighbours per area)
- $\rho_A \in [0, 1)$: spatial correlation in the exposure CAR prior
- σ^2_ϕ: marginal variance of the spatial random effect ϕ_i

As in the thesis proposal, $\mathbf{X}_i$ controls confounding while D_i indexes the measurement error mechanism. D_i may overlap with $\mathbf{X}_i$ but is conceptually distinct.

3 Key Assumptions

The simulation satisfies the following identification assumptions from the thesis proposal.

- **Non-differential surrogate:** $Y_i \perp Z_i \mid A_i, \mathbf{X}_i, D_i$. Once we condition on the true exposure and covariates, the surrogate carries no additional information about the outcome. Satisfied by construction since Y_i is generated from A_i only and Z_i never appears in the outcome equation.
- **Transportability:** The error relationship is assumed transportable to the target population. When direct calibration data are unavailable, inference about A_i is more model-dependent; this is examined in sensitivity scenario S2.

4 Base Model

4.1 Spatial Structure and CAR Prior

Areas are arranged on a $\sqrt{I} \times \sqrt{I}$ synthetic grid with rook contiguity (shared edges only). For this simulation the base value is $I = 400$ (a 20×20 grid); $I = 196$ and $I = 900$ are used as smaller and larger cases.

Following thesis proposal Section 4.3, spatial dependence enters only through the latent exposure model via ϕ_i. The spatial random effect follows the Conditional Autoregressive (CAR) prior:

$$\boldsymbol{\phi} \mid \rho_A, \sigma_\phi^2 \sim \mathcal{N}\big(\mathbf{0},\ \sigma_\phi^2(\mathbf{D}_W - \rho_A \mathbf{W})^{-1}\big)\,. \tag{1}$$

- ϕ_i is a spatially structured random effect: neighbouring areas have similar ϕ_i values, inducing correlation in their true exposures A_i
- The precision matrix $(\mathbf{D}_W - \rho_A \mathbf{W})$ is sparse with adjacent pairs having off-diagonal entry $-\rho_A$ and non-adjacent pairs having zero
- $\rho_A = 0$: areas are spatially independent; $\rho_A \to 1$: strong smoothing across neighbours
- Spatial correlation in the outcome is induced indirectly through A_i; no separate outcome random effect is added

4.2 Covariates and Population Sizes

- **Area confounders:** $X_{i1}, X_{i3}, X_{i4} \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$; $X_{i2} = (G-2)/\sqrt{2}$ with $G \sim \text{Gamma}(2,1)$ is right-skewed (skewness 1.41). All four have mean 0 and variance 1, so the exposure-variance bookkeeping below is unchanged. The skewed one is the confounder with the smallest outcome coefficient, which limits how much its heavy tail inflates the gap between a dataset's covariate-averaged ERF and the population value
- **Population sizes:** $\log(N_i) \sim \mathcal{N}(8.5, 1.2^2)$, used only as the Poisson offset (they do not enter the estimand), giving median $\approx$ 4,900, 5th percentile $\approx$ 600, 95th percentile $\approx$ 40,000. Motivated by Statistics Canada 2021 Census: Ontario has $\approx$ 500 FSAs with total population 14,223,942, ranging from dense urban FSAs in Toronto to sparse rural FSAs in Northern Ontario.

4.3 Calibration Covariate D_i

D_i captures area characteristics that determine prediction error in Z_i.

Base model: scalar D_i. We generate a raw covariate $D_i^* \overset{\text{iid}}{\sim} \text{Gamma}(2, 1)$ (mean 2, variance 2) representing distance to the nearest monitoring station, then standardise:

$$D_i = \frac{D_i^* - 2}{\sqrt{2}},$$

so that D_i has mean 0 and variance 1. Standardisation prevents numerical blow-up in the exponential error variance. The interpretation of η_1 demonstrates that a one standard deviation increase in monitor distance multiplies the error variance by $\exp(\eta_1)$.

- Distance is strictly positive and right-skewed and hence a Gamma distribution naturally showcases this. Most Ontario FSAs are moderately close to one of the 34 monitoring stations, but some rural areas are very far.
- **Motivation:** Gryparis et al. (2009) and Szpiro et al. (2011) show that prediction error variance is primarily determined by distance to the monitor network. MELONS (Evangelopoulos et al., 2026) identified distance, deprivation, and housing type as empirical determinants of $PM_{2.5}$ measurement error in London, UK

Extension: vector D_i. In a full implementation D_i would be a vector with components such as: $D_{i1} \sim \text{Gamma}(2, 1)$ (distance to monitor, standardised), $D_{i2} \sim \mathcal{N}(0, 1)$ (income and employment indicators as a standardised score), $D_{i3} \sim \text{Beta}(2, 2)$ (proportion detached housing). The error model would be $\omega^2(\mathbf{D}_i) = \exp(\eta_0 + \eta_1 D_{i1} + \eta_2 D_{i3})$. Left as a future extension.

4.4 Latent True Exposure

The latent true exposure is generated from:

$$A_i \mid \mathbf{X}_i, \phi_i \sim \mathcal{N}(10 - 0.75X_{i1} + 0.75X_{i2} - 0.75X_{i3} + 0.75X_{i4} + \phi_i,\ 2.25)\,, \tag{2}$$

with $\sigma_A = 1.5$, where ϕ_i is the spatial random effect following the CAR prior defined in equation (1) with parameters (ρ_A, σ_ϕ^2). The spatial random effect ϕ_i induces correlation in A_i across neighbouring areas: areas with ρ_A close to 1 have exposures that vary smoothly across the grid, reflecting a regional wildfire plume, while $\rho_A = 0$ gives spatially independent exposures. The exposure is not truncated: with a mean of 10 and a total SD of about 2.4, the Normal model puts essentially no mass below zero (no negative A_i in 60 test datasets).

- **Mean = 10 μg/m^3:** A cumulative wildfire-smoke exposure level, chosen high enough relative to the spread that A_i and Z_i are essentially never negative, so no truncation or censoring is needed (about 5 to 15 μg/m^3 covers the 1st to 99th percentile)
- $\sigma_A = 1.5$**:** Residual SD; with the covariate and spatial components the total SD of A_i is about 2.4

- **Covariate coefficients ± 0.75:** The signs, set against the outcome coefficients listed under the outcome model below, create real confounding: $\sum_k \gamma_k \beta_k = 0.375$, so an analysis that ignores the confounders has a log-linear exposure slope biased by about 0.07 against a true slope of 0.10 (the confounders' combined effect on the outcome has correlation about 0.2 with A_i). With the sign pattern $(+, -, -, +)$ used by Josey et al. (2023) this sum is exactly zero, so the confounding there operates only through the exposure–covariate interaction
- **Note on skewness:** Ontario $PM_{2.5}$ is heavily right-skewed (skewness ≈ 9.2, 99th percentile ≈ 26.4, extreme values > 200 $\mu g/m^3$ during severe wildfire events). The Normal captures typical days but not extreme events.

Choosing σ_ϕ^2 via R^2 decomposition. With the exposure model now defined, we can determine σ_ϕ^2 by targeting a meaningful proportion of the total variance in A_i. The total variance decomposes into three sources: the covariate contribution $\text{Var}(\mathbf{X}_i \text{ terms}) = 4 \times 0.75^2 = 2.25$, the spatial contribution $\text{Var}(\phi_i)$, and the residual $\sigma_A^2 = 2.25$. Under the CAR prior the average marginal variance of ϕ_i is $\sigma_\phi^2 \, \bar{q}$, where $\bar{q} = \frac{1}{I} \text{tr}\{(\mathbf{D}_W - \rho_A \mathbf{W})^{-1}\}$ is computed exactly from the grid. (The cruder approximation $\bar{q} \approx 1/\bar{d}$ with $\bar{d} \approx 3.6$ neighbours badly understates it at $\rho_A = 0.9$: a target of 0.30 gave a realised R_ϕ^2 of about 0.50.) The proportion of variance explained by the spatial random effect is:

$$R_\phi^2 = \frac{\sigma_\phi^2 \bar{q}}{\text{Var}(A_i)} = \frac{\sigma_\phi^2 \bar{q}}{9 + \sigma_\phi^2 \bar{q} + 9}.$$

Setting a target R_ϕ^2 and solving gives $\sigma_\phi^2 = R_\phi^2 \times 18 / \{(1 - R_\phi^2)\, \bar{q}\}$. At the previously used value of $\sigma_\phi^2 = 0.3$, the implied $R_\phi^2 \approx 0.5\%$ — essentially no spatial contribution. We therefore parameterise through the target R_ϕ^2, with base value 0.20, and compute the implied σ_ϕ^2 accordingly; the realised share is about 0.20.

4.5 Measurement Error Model

Two parameterisations of the measurement error model are considered. Both generate Z_i as a function of the true exposure A_i and calibration covariate D_i, and both satisfy the non-differential assumption by construction since Z_i does not appear in the outcome equation.

In both versions the measurement errors $\varepsilon_i = Z_i - \mathbb{E}[Z_i \mid A_i, D_i]$ are assumed conditionally independent across areas given (A_i, D_i). This is a simplifying assumption: Gryparis et al. (2009) show that prediction errors from spatial models are spatially correlated because nearby areas share the same monitoring station data when their surrogates are generated. Incorporating spatially correlated errors is listed as a potential extension.

Version A: Classical error with Di-dependent mean (primary model, agreed with the supervisor). Motivated by Wu et al. (2019), where covariates D enter the model relating the true and error-prone exposures, D_i enters through the mean of the surrogate, introducing systematic directional bias (note the direction here is surrogate given truth, the opposite of Wu et al.; see the note after this list and Version C):

$$Z_i \mid A_i, D_i \sim \mathcal{N}(\beta_{Z1} A_i + \beta_{Z2} D_i,\ \sigma_Z^2). \tag{3}$$

This model is not fully identifiable when A_i is latent: β_{Z1} and σ_Z^2 cannot be separately recovered from the marginal distribution of Z_i without validation data.

- β_{Z1}: attenuation. $\beta_{Z1} = 1$ centres the surrogate on the truth; $\beta_{Z1} < 1$ attenuates high exposures. (Wu et al. (2019) use $\gamma_1 = 0.8$, but that is the slope of the true exposure on the surrogate, not of the surrogate on the truth, so it is not a default for β_{Z1}.)
- β_{Z2}: directional bias from D_i. When $\beta_{Z2} > 0$ the surrogate overestimates true exposure in high-D_i areas (far from monitors), consistent with the MELONS finding of systematic overestimation in deprived and remote areas. When $\beta_{Z2} = 0$ this reduces to standard classical error.
- σ_Z^2: fixed homoscedastic noise. Heterogeneity across areas is captured entirely through the mean shift $\beta_{Z2} D_i$.

Wu et al. (2019) write their model in the regression calibration direction, $E(X \mid W, D) = \gamma_0 + \gamma_1 W + \gamma_2^\top D$, i.e. the true exposure given the surrogate, and in their simulations they generate the surrogate W first and then $X \mid W, D$. Version A is therefore not their model with renamed coefficients: the two have different structure (see Version C), and their γ cannot be reused as β_Z. Neither Wu et al. nor this document uses the terms classical or Berkson for these models; the labels are ours.

Version B: Classical error with Di-dependent variance (earlier primary model, now an alternative). D_i enters through the variance of the surrogate, relaxing the homoscedasticity assumption of Josey et al. (2023) Assumption 2:

$$Z_i \mid A_i, D_i \sim \mathcal{N}\big(\exp(\beta_1 D_i)\, A_i,\ \sigma_Z^2(D_i)\big), \qquad \sigma_Z^2(D_i) = \exp(\eta_0 + \eta_1 D_i). \tag{4}$$

The surrogate is not truncated; with $\eta_0 = -0.5$ a negative Z_i is very rare (at most a couple of areas in occasional datasets) and is harmless because the Normal error model is exact.

- With $\eta_0 = -0.5$ the base value $\eta_1 = 0.5$ keeps the error SD moderate (median about 0.7, 99th percentile about 1.8, typical maximum about 2.7). At $\eta_1 = 1$ the right tail of D_i would give a handful of areas a far larger error SD, and these areas would carry most of the total error variance.
- The calibration slope $\exp(\beta_1 D_i)$ is close to 1 on average ($\beta_1 = -0.1$ gives a mean slope of about 1.0) but falls below 1 in areas far from monitors (roughly 0.7 to 0.8 for $D_i > 1.5$), so the surrogate is attenuated and also noisier in high-D_i areas. The log link keeps the slope positive despite the long right tail of D_i
- $\eta_1 = \beta_1 = 0$ recovers Josey's homoscedastic, unattenuated Assumption 2 exactly
- $\exp(\eta_0 + \eta_1 D_i)$ ensures positivity and implies a log-linear (proportional) relationship: a unit increase in D_i multiplies the error variance by $\exp(\eta_1)$, natural for distance-based quantities (Gryparis et al., 2009)
- Unlike Version A, there is no directional shift in the mean; the departures from classical error are an attenuation and a noise magnitude that both depend on D_i
- This version is identifiable: the marginal variance of Z_i given $(\mathbf{X}_i, D_i)$ is $\exp(2\beta_1 D_i)\,(\sigma_A^2 + \sigma_\phi^2 \bar{q}) + \exp(\eta_0 + \eta_1 D_i)$, allowing estimation of (η_0, η_1) without observing A_i

Version C: Berkson-direction error with a D_i shift (possible specification, not implemented). Following the direction of Wu et al. (2019), the surrogate is the starting point and the true exposure is generated from it, so the error sits in the truth and is independent of the surrogate:

$$A_i \mid Z_i, D_i \sim \mathcal{N}(b_1 Z_i + b_2 D_i,\ \sigma_e^2). \tag{5}$$

This is Version A with the roles of Z_i and A_i swapped. Here Z_i is an observed covariate and A_i is the latent quantity. The non-differential assumption still holds by construction, since Z_i does not appear in the outcome equation.

- b_1: calibration slope. $b_1 = 1$ means the truth is centred on the surrogate; Wu et al. use $\gamma_1 = 0.8$ in their simulations. There is no intercept, as in Version A, so the surrogate is unbiased on average (and the location of A_i is anchored by Z_i).
- b_2: shift per SD of D_i. When $b_2 > 0$ the truth is higher than the surrogate in high-D_i areas (far from monitors), so the surrogate underestimates exposure there. This is the same kind of directional bias as β_{Z2} in Version A, seen from the other side.
- σ_e^2: the variance of the truth that the surrogate and D_i do not explain.
- **Simulation:** to keep the confounding and the outcome identical to Versions A and B, the surrogate is built from the covariate and spatial part of the exposure plus noise, $Z_i = (\mu_i + \phi_i)/b_1 + w_i$ with $\mu_i = \gamma_0 + \sum_k \gamma_k X_{ik}$ and $w_i \sim \mathcal{N}(0, \sigma_w^2)$, and then A_i is drawn from equation (5). Setting $\sigma_w^2 = (\sigma_A^2 - b_2^2 - \sigma_e^2)/b_1^2$ keeps $\mathrm{Var}(A_i \mid \mathbf{X}_i, \phi_i) = \sigma_A^2 = 2.25$. Base values: $b_1 = 1$, $b_2 = 0.5$, $\sigma_e = 0.75$

In a test dataset with these values the regression of A on (Z, D) recovered slope 1.00, D coefficient 0.48 and residual SD 0.72, the reliability $\mathrm{corr}^2(A, Z)$ was 0.86 (about 0.83 for Version A), and a naive log-linear analysis on Z gave a slope of 0.101 against a true 0.10. So, unlike under Versions A and B, a naive analysis is essentially unbiased for the slope here; it is biased for curved exposure–response functions, since $E[f(A) \mid Z] \neq f(Z)$. This makes Version C a check that a correction does no harm when the naive analysis is nearly right.

Reversing the direction does not resolve identifiability. A_i is never observed, so its scale is not fixed by the data: (b_1, b_2, σ_e) can be rescaled together with the outcome slope without changing the distribution of (Y_i, Z_i, D_i). Leaving out the intercept anchors the location, and fixing $b_1 = 1$ anchors the scale. That leaves b_2 and σ_e identified only through the outcome, using the assumption that D_i affects the outcome only through A_i. Wu et al. avoid this with a validation sample in which A_i is observed. Version C is documented as an alternative only; it is not implemented in `data.generate()` and is not planned for the main study.

4.6 Outcome Model

The area-level count outcome follows a Poisson model with a log-linear predictor and population offset, matching Josey et al. (2023):

$$Y_i \mid A_i, \mathbf{X}_i, N_i \sim \mathrm{Poisson}(N_i \exp(\eta_i)), \tag{6}$$

$$\eta_i = -3 + f(A_i) - 0.5 X_{i1} - 0.25 X_{i2} + 0.25 X_{i3} + 0.5 X_{i4}. \tag{7}$$

- **Population offset** $\log(N_i)$**:** N_i scales the expected count so that $\exp(\eta_i)$ is the event rate per person. A larger area will have more events simply because it has more people; the offset removes this confound. In R this enters as an additive term $\log(N_i)$ in the linear predictor or via the `offset` argument.
- **Intercept** -3**:** $\exp(-3) \approx 5\%$ baseline event rate, representing baseline depression incidence in the CLSA population (Raina et al., 2019)
- **Confounder coefficients** $\pm(0.5, 0.25)$**:** Follow Josey et al. (2023)
- **ERF** $f(A_i)$**:** Exposure-response function specified in sensitivity scenario S4; three functional forms are evaluated
- **No spatial random effect:** Consistent with thesis proposal Section 4.3; spatial correlation in outcomes is induced indirectly through the spatially correlated A_i surface

Z_i does not appear in equation (6), satisfying the non-differential assumption by construction.

4.7 Causal Estimand

Following Josey et al. (2023), the target estimand is the marginal exposure-response function (ERF):

$$\theta(a) \equiv \mathbb{E}[Y_i(a)/N_i],$$

the average event rate per person across areas, where $Y_i(a)$ is the count that would occur in area i if, possibly contrary to what is observed, it had received exposure level $A_i = a$. Every area counts equally; expected counts are recovered by multiplying the rate by N_i. The causal contrast between two exposure levels is $\Delta(a_1, a_0) = \theta(a_1) - \theta(a_0)$. Under the identifying assumptions in Section 3, the ERF is identified by the g-formula:

$$\theta(a) = \mathbb{E}[\mathbb{E}[Y_i/N_i \mid A_i = a, \mathbf{X}_i]] = \mathbb{E}\Big[\exp(-3 + f(a, X_{i1}) + \mathbf{X}_i^\top \boldsymbol{\beta})\Big].$$

- **Truth:** $\theta(a)$ is the mean of $\exp(-3 + f(a, X_{i1}) + \mathbf{X}_i^\top \boldsymbol{\beta})$ over areas, with no N_i weights. Coverage is reported against the *population* truth, the mean over 2×10^6 simulated areas drawn from the same covariate distribution (identical in every replicate), and also against each dataset's *sample* truth (the mean over that dataset's own 400 areas, as in Josey et al. 2023)
- **Estimators:** g-computation averages the fitted rate with Dirichlet$(1, \ldots, 1)$ Bayesian-bootstrap weights over areas, one weight vector per posterior draw
- **Evaluation grid:** the integers 5 to 15 (11 values), covering the bulk of the A_i distribution (1st to 99th percentile about 4.6 to 15.4)
- **Primary causal contrast: extreme vs typical exposure:** $\Delta(Q_{99}, Q_{50})$ where Q_{99} and Q_{50} are the 99th and 50th percentiles of the simulated A_i distribution. For the base model these correspond approximately to $Q_{50} \approx 10$ and $Q_{99} \approx 15.4$, representing the increase in the average depression rate on a severe wildfire day relative to a typical day. The percentiles are computed from the simulated A_i values in each replicate so that the contrast automatically adapts to the exposure distribution under each scenario.
- **Secondary contrast: IQR:** $\Delta(Q_{75}, Q_{25})$ following standard practice in air pollution epidemiology

5 Simulation Parameter Grid

Table 1: Simulation parameters. Version A and Version B refer to the two error model parameterisations in Section 4. R^2_ϕ denotes the proportion of $\text{Var}(A_i)$ explained by the spatial random effect.

Parameter	Notation	Base / Range	Justification
Design			
Number of areas	I	400	20×20 grid; Josey et al. (2023) use 400 and 800 clusters
Replicates	R	200	
ERF evaluation points		11 (integers 5 to 15)	Covers the 1st to 99th percentile of A_i
Covariates			
Area confounders	X_{ik}	X_{i2} standardised Gamma(2,1); others Normal	Mean 0, variance 1
Population size	$\log(N_i)$	$\mathcal{N}(8.5, 1.44)$	Statistics Canada 2021 Census
Calibration covariate	D_i	Gamma$(2, 1)$, standardised	Monitor distance proxy; Gryparis et al. (2009)
True exposure			
Mean	$\mu_{A,i}$	$10 + 0.75(-X_{i1} + X_{i2} - X_{i3} + X_{i4})$	Mean 10 μg/m^3; keeps A_i and Z_i non-negative without truncation
SD	σ_A	1.5	Residual SD; total SD of A_i about 2.4
Spatial R^2	R^2_ϕ	0.20	Proportion of $\text{Var}(A_i)$ explained by spatial effect (exact CAR calibration)
Spatial correlation	ρ_A	0.9	Base; $\rho_A = 0$ used only as a no-spatial check
Error model Version A			
Attenuation	β_{Z1}	1	Surrogate centred on the truth; < 1 attenuates
Directional bias	β_{Z2}	0.5	Shift per SD of D_i; 0 = no bias, > 0 = overestimation in high-D_i areas
Noise variance	σ^2_Z	1	Fixed moderate noise; reliability $\text{Var}(A)/\text{Var}(Z) \approx 0.83$
Error model Version B			
Baseline variance	η_0	-0.5	$\exp(-0.5) \approx 0.61$ at $D_i = 0$; reliability $\text{Var}(A)/\text{Var}(Z) \approx 0.77$
Di-slope	η_1	0.5	Moderate heteroscedasticity; $\eta_1 = 0$ recovers Josey
Attenuation slope	β_1	-0.1	Calibration slope $\exp(\beta_1 D_i)$: about 1 on average, roughly 0.7 to 0.8 far from monitors
Outcome			
Intercept		-3	$\approx 5\%$ baseline event rate (CLSA)
ERF	$f(A_i)$	Linear/Quadratic/Nonlinear	
Confounder coefs	$\boldsymbol{\gamma}$	$(-0.5, -0.25, 0.25, 0.5)$	
Sensitivity scenarios			
D_i^* distribution		Exp(1); empirical	Monotone decreasing; real Ontario data
N_i distribution		Uniform$(10, 1000)$	Josey comparability
Exposure distribution		Log-normal	Extreme wildfire events
Number of areas	I	$\{196, 900\}$	Scaling; all perfect squares

6 Sensitivity Scenarios

D_i **distribution.** **Gamma(2, 0.5)**: mean 4, more heterogeneity between areas. **Exponential(1)**: monotone decreasing, consistent with Southern Ontario station concentration. **Empirical**: actual FSA-to-NAPS distances from Statistics Canada shapefiles.

Error model alternatives. **Version A vs Version B**: Data generated under Version A (directional bias, $\beta_{Z2} \in \{0, 1, 2\}$) but analysed with a Version B error model. Quantifies performance degradation when the true error has directional bias that the proposed method does not model. $\beta_{Z2} = 0$ provides an internal consistency check.

Spatial correlation. **Scenario B ($\rho_A = 0$)**: spatially independent exposures. Directly comparable to Josey et al. (2023) which sets $\phi_i = 0$ for all areas. **Scenario A ($\rho_A = 0.9$)**: high spatial smoothing, reflecting a sustained regional wildfire event where nearby FSAs share similar smoke concentrations.

Exposure-response functional forms. Three forms spanning simple to complex, crossed with spatial scenarios to give six main simulation cells.

Linear: $f(A_i) = 0.10(A_i - 10)$. Monotone increasing with no curvature.

Quadratic: $f(A_i) = 0.30(A_i - 10) - 0.05(A_i - 10)^2$. Concave shape that rises then declines within the evaluation window (vertex at $A_i = 13$).

Nonlinear (up–down–up): $f(A_i, X_{i1}) = 0.12(A_i - 10) + 0.55\big[\sin\big(2\pi(A_i - 6.25)/9\big) - 0.5\big] - 0.25\tanh\big((A_i - 10)/4\big)X_{i1}$, centred so that $f(10, \cdot) = 0$ like the other forms. A linear trend plus one sine wave: the curve rises to a local maximum near $A_i = 9$, falls to a local minimum near $A_i = 12.5$, and then rises again through the end of the window. It is not captured by a linear or quadratic form, and it grows only linearly beyond the window (exposure rarely exceeds 17), unlike a cubic. The bounded interaction with X_{i1} (at most 0.25 per SD) makes the exposure effect depend on X_{i1}, in the spirit of Josey et al. (2023) but without their unbounded $(A_i - 10)X_{i1}$ term; the marginal ERF is then a genuine average of different conditional curves.

Log-normal exposure. $\log(A_i) \sim \mathcal{N}(1.68, 0.54)$ matching Ontario moments: median 5.4, mean 7, SD 5.9. Captures extreme wildfire events (99th percentile ≈ 26 μg/m^3) that the Normal cannot represent.

7 Performance Metrics

For each scenario and method (naive, oracle, proposed, regression calibration):

- **Bias**: $\hat{\theta}(a) - \theta(a)$ averaged over evaluation points and replicates
- **RMSE**: $\sqrt{R^{-1}\sum_r(\hat{\theta}_r(a) - \theta(a))^2}$
- **Coverage**: fraction of 95% credible intervals containing $\theta(a)$

- **Interval width**: mean width of 95% credible intervals
- **Causal contrast bias**: $\hat{\Delta}(Q_{99}, Q_{50}) - \Delta(Q_{99}, Q_{50})$ and $\hat{\Delta}(Q_{75}, Q_{25}) - \Delta(Q_{75}, Q_{25})$, where quantiles are computed from the simulated A_i distribution in each replicate

8 Potential Extensions

- **Spatially correlated measurement errors:** The current model assumes conditional independence of measurement errors $\varepsilon_i = Z_i - \mathbb{E}[Z_i \mid A_i, D_i]$ across areas which is a simplifying assumption that Gryparis et al. (2009) show is violated in practice: prediction errors from spatial models are correlated across nearby areas because those predictions are generated using the same monitoring station data. A natural extension is to give the error term its own spatial covariance structure, either by making D_i spatially structured via a second CAR prior (so that nearby areas have correlated error variances), or by adding an explicit spatially correlated residual $\psi_i \sim \text{CAR}(\rho_\varepsilon, \sigma_\varepsilon^2)$ to the error model.
- **Individual-level binary outcome:** Replacing the area-level Poisson count with individual-level Bernoulli outcomes $Y_{ij} \sim \text{Bernoulli}(p_{ij})$ and individual covariates L_{ij}, matching the CLSA data structure. The Bayesian bootstrap would draw Dirichlet weights over all $N = \sum_i n_i$ individuals rather than over areas.
- **Spatial random effect on outcome (ν_i)**: A second CAR prior capturing residual spatial clustering in health outcomes (Englert et al., 2025).
- **Negative binomial outcome:** The Poisson model assumes $\text{Var}(Y_i) = \mathbb{E}[Y_i]$, which is often violated in practice when area-level counts are overdispersed due to unmeasured heterogeneity across FSAs. A negative binomial outcome relaxes this by introducing a dispersion parameter $\xi > 0$:

  $$Y_i \mid A_i, \mathbf{X}_i, N_i \sim \text{NegBin}(\mu_i, \xi), \qquad \mu_i = N_i \exp(\eta_i),$$

  As $\xi \to \infty$ this recovers the Poisson; small ξ gives strong overdispersion. The dispersion parameter $\xi = 1$ is a natural starting point and is the value used by Englert et al. (2025) in their simulation. This extension tests whether the proposed method remains well-calibrated when the outcome model is more variable than the base Poisson.
- **Zero-inflated outcome**: For rare outcomes or small populations where structural zeros arise.
- **Log-normal exposure model:** The base model specifies A_i on the original scale as a Normal with a floor at zero. An alternative motivated by the right-skewed nature of $PM_{2.5}$ concentrations and by the suggestion that $\log(A_i)$ should be regressed on covariates is to model the log-exposure directly:

  $$\log(A_i) \mid \mathbf{X}_i, \phi_i \sim \mathcal{N}\big(\gamma_0 + \gamma_1 X_{i1} + \gamma_2 X_{i2} + \gamma_3 X_{i3} + \gamma_4 X_{i4} + \phi_i,\ \sigma^2_{\log A}\big)\,,$$

  so that $A_i = \exp(\log A_i) > 0$ always. The parameters are matched to Ontario $PM_{2.5}$ empirical moments: setting $\gamma_0 = 1.68$ and $\sigma_{\log A} = 0.73$ gives $\mathbb{E}[A_i] \approx 7\ \mu\text{g/m}^3$ and $\text{SD}(A_i) \approx 5.9\ \mu\text{g/m}^3$, matching the provincial monitoring data. The error model in Version B would then operate on the log scale as $\log(Z_i) \mid \log(A_i), D_i \sim \mathcal{N}(\log(A_i), \exp(\eta_0 + \eta_1 D_i))$, giving multiplicative error on the original scale.